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NE 5350 – Nuclear Criticality Safety Homework 1

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1 Mayak 1958 Criticality Accident

Using the concepts of exponential power growth and reactivity feedback discussed in class, describe the Mayak 1958 criticality accident.

The uranyl nitrate solution, enriched at 90%, was sub-critical in the test tank before the accident. The tank belonged to a facility built to measure the critical parameters of highly enriched uranium solutions, so it was not a safe-geometry vessel by design; instead, the amount of solution and its shape in the upright tank kept the system sub-critical. In this condition the reactivity is static, meaning that ρ is constant in time, and

$$\rho = \frac{k_{\text{eff}} - 1}{k_{\text{eff}}} < 0. \quad (1)$$

Reactivity determines how the neutron population changes in time. For $\rho < 0$ the population decays or, in the presence of an external neutron source as in this facility, settles at a constant sub-critical multiplication level. For $\rho > 0$ the population grows exponentially. Reactivity feedback acts in the other direction: the neutron population is proportional to the fission rate, and therefore to power, so a growing population deposits energy in the solution. The resulting heating, thermal expansion, radiolytic gas generation, and expulsion of solution change ρ . In fissile solutions this feedback is negative, so it opposes the growth of the neutron population.

After completing an experiment, the experimentalists needed to drain the solution from the test tank into several safe-geometry tanks. To speed up the draining, three of them unbolted the test tank from its mounting and tipped it. As the tank tipped, the solution collected on one side and more-or-less resembled the geometry of a sphere (??), which reduces neutron leakage. At the same time, the bodies of the experimentalists reflected neutrons back into the solution. The combination of the solution's geometry and the neutron reflection from the experimentalists' bodies inserted positive reactivity, so that $\rho > 0$ and the neutron population grew exponentially.



Figure 1: From Solution Sitting Flat in Cylinder to Becoming Sphere-like

The excursion was a single power pulse of roughly 2.3×10^{17} fissions. As the power rose, part of the solution was ejected from the tank. The ejection changed the geometry of the solution and removed fissile material from the compact configuration, which is a large negative reactivity feedback that terminated the pulse. Three of the four experimentalists died within about a week, and the fourth, who was about 3 m from the tank, developed acute radiation syndrome.

Consider the production and destruction of neutrons and delayed neutron precursors (DNPs), which have been provided during the lecture, as an approximation of a zero-dimensional system:

$$\frac{d}{dt}n(t) = \frac{\rho(t) - \beta}{\Lambda}n(t) + \sum_{d=1}^{N_D} \lambda_d C_d(t) \quad (2)$$

$$\frac{d}{dt}C_d(t) = \frac{\beta_d}{\Lambda}n(t) - \lambda_d C_d(t) \quad (3)$$

Where $n(t)$ is the neutron population, ρ is reactivity, β_d is the effective delayed neutron fraction of precursor group d , $\beta = \sum_{d=1}^{N_D} \beta_d$ is the total effective delayed neutron fraction, λ_d is the decay constant [s^{-1}], Λ is the mean neutron generation time, and C_d is the concentration of DNP group d . With these terms defined, the equations can be written in matrix form:

$$A(t) = \begin{pmatrix} \frac{\rho(t) - \beta}{\Lambda} & \lambda_1 & \lambda_2 & \cdots & \lambda_{N_D} \\ \frac{\beta_1}{\Lambda} & -\lambda_1 & & & \\ \frac{\beta_2}{\Lambda} & & -\lambda_2 & & \\ \vdots & & & \ddots & \\ \frac{\beta_{N_D}}{\Lambda} & & & & -\lambda_{N_D} \end{pmatrix} \quad (4)$$

$$\vec{x}(t) = \begin{pmatrix} n(t) \\ C_1(t) \\ \vdots \\ C_{N_D}(t) \end{pmatrix} \quad (5)$$

$$\frac{d}{dt}\vec{x}(t) = A(t)\vec{x}(t) \quad (6)$$

This system is linear in $\vec{x}$ for any $A(t)$. Following a reactivity insertion that holds ρ constant, A is constant and the solution is

$$\vec{x}(t) = \exp(At) \vec{x}_0 \quad (7)$$

such that $\exp$ is a matrix exponential. The growth rate is set by the dominant eigenvalue of A . For $\rho < 0$ this eigenvalue is negative and the population decays. For $0 < \rho < \beta$ it is positive and the growth is limited by the delayed neutrons. For $\rho > \beta$ the system is prompt critical and the population grows on the time scale of the prompt neutron generation time Λ . Reactivity feedback makes ρ depend on the state, $\rho = \rho(\vec{x})$, so A depends on $\vec{x}$, the system becomes nonlinear, and ?? no longer applies. In the Mayak accident, this negative feedback and the ejection of solution reduced ρ and ended the exponential growth.

2 Demon Core 1 and 2

In the first “Demon Core” criticality accident, the operator remained in the room after the accident thereby significantly increasing the dose that he received. In the second “Demon Core” criticality accident, the operator described physically separating the two hemi-spheres though the power excursion had already been terminated by reactivity feedback effects. For both of these criticality safety accidents, describe how the physics of reactivity feedback could have been used to modify the experiments to make them safer.

Tables

III Acronyms